

Standard Operating Procedure ***Miter Saw***

Ridgid Miter Saw

The Ridgid Miter Saw is a high-torque, precision tool designed for cross-cutting, mitering, and beveling. It features a **15-amp motor** delivering **5,000**. This machine is a roughing and semi-finishing tool; it is not intended for ripping lumber.

Potential Hazards

- **Kickback**: Occurs if the workpiece binds or is not flat against the fence, causing the wood to fold and throw the operator's hand toward the blade.
- **Mechanical**: High-speed rotating blade ($5,000$ RPM) and sliding carriage "pinch points".
- **Projectile**: Small off-cuts (shrapnel) can be launched if the saw arm is raised while the blade is still spinning.
- **Electrical**: Potential for unexpected startup if the trigger safety is bypassed or damaged

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	**ANSI Z87.1-compliant safety goggles:** Mandatory. Mandatory with side shields. The high RPM and blade speed (120 mph at the tip) create high-velocity debris.
Face Protection	**Face Shield:** Highly recommended in addition to goggles to protect against "wheel burst" and heavy sparks.



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PPE Item	Requirement
No Gloves Advised	**Strictly Forbidden:** Gloves are an entanglement hazard. If a glove snags a carbide tooth, it will pull the operator's hand into the blade before the motor can stall.
Hearing Protection	**Required:** Operating noise levels under load exceed 90 dB; earplugs or muffs are mandatory.
Apparel & Hair	Tie back long hair and secure loose clothing. Remove all jewelry, especially rings and watches.
Footwear	Appropriate enclosed footwear must be worn at all times.

Before Use

- **Blade Inspection**: Ensure the carbide-tipped blade is sharp and the arbor nut is torqued. Never use a blade with missing teeth.
- **Guard Integrity**: Verify the **lower blade guard** retracts smoothly and covers the blade completely when the arm is raised.
- **Fence Check**: Ensure the sliding miter fences are adjusted to provide support for the stock but are clear of the blade's path at both 0 and 45 bevels.

During Use

- **Startup**: Allow the motor to reach full set speed prior to engaging the stock.
- **Workholding**: **Never cut freehand.** All stock must be secured in the integrated quick-release vise.
- **The "4-Inch" Rule**: Keep hands at a safe distance of at least 4 inches from the abrasive wheel at all times.



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- **Cutting Logic**:
    - Do not force the blade into the stock; let the RPM of the saw perform the cut.
    - Maintain steady, even pressure to prevent wheel glazing or motor stall.
- **Spark Deflection**: Ensure the adjustable spark deflector is aimed away from other personnel and flammable waste.
- **The Stop Rule**: Wait for the abrasive wheel to come to a **complete stop** before lifting the saw head or attempting to remove the workpiece.

After Use

- **Power Down**: Release the trigger and ensure the machine is ready for the next user.
- **Coast-Down Safety**: Wait for the blade to stop on its own; do not attempt to stop it manually.

- **Debris Removal**: Use a brush or shop vacuum to collect metal dust and shards once the work area is cool.

- **Warning**: Avoid cleaning with compressed air to prevent eye injury from metal fines.
- **Storage**: Lower the saw head and engage the lock-down pin to secure the machine.
- **Personal Hygiene**: Wash hands and exposed skin thoroughly after work to remove metallic dust and grit.

Emergency Equipment and Procedures

- **Emergency Stop**: Immediately release the trigger and disconnect the power if a malfunction occurs.



- **Blade/Wheel Failure**: If the wheel fragments, keep the saw head **DOWN** and wait for the motor to stop before investigating.
- **Reporting**: Report all incidents, injuries, or near-misses to the Laboratory Project Supervisor immediately.